

Alayna McInerny

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EDUCATION

University of Minnesota

Bachelor of Electrical Engineering, Minor in Computer Science

Minneapolis, MN

Aug 2024 – May 2028

EXPERIENCE

Trane Technologies

Software Engineering Intern

May 2026 – Present

St. Paul, MN

- Developed a desktop GUI application using PySide6 to generate controller configuration files, saving technicians up to 1 hour per job site and allowing for on-site testing
- Integrated GUI with Tranes TracerTU platform, successfully migrating desktop function to a web-based architecture, closely following company privacy standards

STRIVE Medical

President

May 2026 – Present

Minneapolis, MN

- Work with all sub-team members to ensure product follows set timeline, communicate between groups
- Host professional events with professors in the medical field
- Drive campus recruitment by presenting to engineering classes, managing weekly meetings for over 40 members, and coordinating projects with external organizations

University of Minnesota

Undergraduate Researcher

Sep. 2025 – Present

Minneapolis, MN

- Utilize Altium Designer to design custom PCBs
- Complete hardware in the loop testing on high voltage power electronics to determine system reliability and safety
- Present research monthly and summarize work done for newsletters

Indiana University

Teaching Assistant

Jan. 2025 – May 2025

Bloomington, IN

- Help lead both lectures and labs for class of 30 students
- Host office hours several times per week to connect with students outside of class time
- Attended special training to gain access to laser cutters, helped students use special equipment

Society of Women Engineers

Treasurer

Sep. 2024 – May 2025

Bloomington, IN

- Managed financial budgeting and fundraising efforts while collaborating with cross-functional campus organizations to host school-wide events

PROJECTS

SynDex | C++, C, Altium

Oct. 2025 – May 2026

- Developed a multi-axis surgical training robot, working with electrical, mechanical, firmware, and software teams
- Designed and embedded firmware to interface with I2C encoders and ODrive motor controllers for instant haptic feedback that mimicked the pressure of human tissue
- Presented prototype at the Mostly Viable Project competition, placing first and receiving funding for future projects
- Researched and implemented admittance control algorithm to ensure smoother movement from the user

Animatronic Eyes | Assembly, C, C++

Spring 2026

- Developed assembly code for two Servo motors
- Detected light levels using a photo resistor, high light levels activated all motors and triggered movement for animatronic eyeballs
- Printed and assembled a LCD 7-Segment display

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL, HTML/CSS, Verilog

Developer Tools: Git, VS Code, Visual Studio, IntelliJ, Eclipse, Linux

Software: Altium, Matlab/MatlabSimulink, Solidworks, Fusion360

Concepts: HIL Testing, PCB Design, Embedded Systems, Signal Processing, Root Cause Analysis